

LIBERAL, KS, USA

WMO: 724516

Lat: 37.050N

Lon: 100.967W

Elev: 2873

StdP: 13.23

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23020

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.0	11.6	-0.2	6.0	13.7	2.9	7.1	18.2	34.7	44.6	31.2	41.4	10.4	20	0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	100.2	69.4	98.8	69.4	95.8	69.0	74.2	89.5	73.0	89.1	71.9	88.3	16.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.4	124.8	78.6	69.6	121.0	78.1	67.0	110.4	77.2	39.7	88.7	38.6	89.5	37.5	88.3	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
32.3	28.1	25.3	-1.4	105.2	4.8	2.5	-4.8	107.0	-7.6	108.5	-10.3	109.9	-13.7	111.8
			-3.0	75.8	4.1	1.1	-6.0	76.6	-8.4	77.3	-10.7	77.9	-13.7	78.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	34.4	37.1	46.0	54.8	65.0	75.5	80.2	70.6	57.2	44.5	34.4		
		DBStd	18.67	9.81	11.00	10.61	9.54	8.50	6.51	5.13	5.36	7.99	9.30	9.09		
		HDD50	1857	488	374	198	58	4	0	0	1	37	211	486		
		HDD65	4634	949	780	591	323	108	6	0	1	40	272	616		
		CDD50	4264	3	15	74	203	468	765	935	873	619	261	45		
		CDD65	1566	0	0	3	18	107	320	471	409	208	30	0		
		CDH74	19688	1	17	129	494	1536	3915	5913	4707	2404	531	41		
		CDH80	10331	0	2	27	170	680	2078	3425	2599	1157	189	4		
(14)	Wind	WSAvg	12.3	11.2	12.0	13.2	14.4	13.3	13.5	11.8	11.0	12.1	12.2	11.8	11.2	
(15)	Precipitation	PrecAvg	19.6	0.5	0.6	1.3	1.5	2.8	3.0	2.9	2.3	1.9	1.5	0.9	0.5	
(16)		PrecMax	34.1	2.0	2.3	9.0	4.3	8.8	8.1	11.9	6.8	6.4	10.6	3.6	1.9	
(17)		PrecMin	12.1	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.1	0.0	0.0	0.0	0.0	
(18)		PrecStd	4.5	0.4	0.6	1.6	1.1	1.9	1.9	2.0	1.5	1.5	1.9	0.9	0.5	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	70.3	75.5	83.6	90.5	97.5	104.0	103.6	102.1	99.5	90.9	79.3	68.4		
(20)		0.4% MCWB	49.3	50.8	56.5	58.5	63.5	66.8	70.8	69.8	67.7	62.6	54.6	49.9		
(21)		2% DB	63.5	69.8	78.1	84.4	91.4	99.0	100.2	99.6	95.0	85.2	72.7	61.4		
(22)		2% MCWB	46.1	49.4	54.7	56.9	63.6	68.1	70.2	70.0	67.0	61.5	53.0	46.0		
(23)		5% DB	57.1	63.0	71.9	80.6	88.1	95.2	99.0	97.1	90.9	80.7	66.5	55.6		
(24)		5% MCWB	43.5	46.9	52.6	57.6	63.6	68.5	70.2	69.9	67.0	60.0	50.4	44.1		
(25)		10% DB	51.9	55.4	65.8	73.5	82.4	91.2	96.5	93.0	87.7	74.5	61.4	51.5		
(26)		10% MCWB	41.1	43.0	50.6	55.0	62.6	68.4	70.2	69.7	66.4	58.0	47.8	41.3		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	50.9	54.8	61.8	64.8	70.8	75.1	76.4	75.2	72.2	68.2	58.5	53.0		
(28)		0.4% MCDB	66.3	67.6	75.1	75.4	84.0	87.8	90.5	90.1	87.7	79.9	70.4	61.3		
(29)		2% WB	47.4	50.8	57.8	61.8	68.7	73.2	74.6	73.5	70.5	65.6	54.8	48.5		
(30)		2% MCDB	61.8	66.8	69.8	75.3	81.6	88.3	89.7	90.0	86.4	75.3	67.9	60.4		
(31)		5% WB	44.6	47.9	54.9	59.3	66.9	71.7	73.3	72.4	69.2	63.3	52.2	45.0		
(32)		5% MCDB	56.7	62.7	67.9	74.7	80.4	87.8	89.6	88.9	85.3	74.0	65.6	55.2		
(33)		10% WB	41.7	44.7	52.0	56.9	64.9	70.3	72.1	71.2	67.9	60.4	49.5	41.9		
(34)		10% MCDB	51.4	55.5	66.0	72.5	78.6	86.2	88.9	87.5	82.9	72.6	60.6	51.2		
(35)	Mean Daily Temperature Range	MDBR	26.4	27.2	28.6	28.1	26.9	26.3	26.4	25.3	26.3	27.2	27.9	25.0		
(36)		5% DB MCDBR	37.0	38.6	38.6	38.0	34.3	32.2	30.8	31.1	31.9	35.3	36.9	34.0		
(37)		5% WB MCWBR	22.4	22.1	20.0	17.3	13.9	11.0	9.3	9.4	11.2	16.2	20.1	21.0		
(38)		5% WB MCDBR	35.5	36.5	33.2	31.8	29.0	27.7	26.6	26.4	26.9	27.6	33.4	32.5		
(39)	Clear-Sky Solar Irradiance	taub	0.259	0.267	0.298	0.327	0.345	0.374	0.384	0.368	0.338	0.303	0.272	0.258		
(40)		taud	2.569	2.561	2.474	2.414	2.430	2.392	2.395	2.408	2.474	2.543	2.565	2.572		
(41)		Ebn at Noon	303	312	308	303	297	287	283	286	290	293	294	296		
(42)		Edh at Noon	23	27	32	36	36	38	37	36	32	27	23	22		
(44)	All-Sky Solar Radiation	RadAvg	884	1142	1538	1903	2184	2363	2309	2032	1722	1278	967	779		
(45)		RadStd	78	117	131	108	134	124	99	102	88	146	88	57		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+1.12	+1.57	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page